



Computer Networks

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Agenda (1st Part)

- Intro
- What's a computer network
- Computer Network Components
- Device Types
- Protocols
- Topologies
- Reliable Network Characteristics
- Internet of Things



Computer System & Networks



TCP/IP Protocol Stack

Link Layer/Ethernet

IP Networks

TCP

Application Layer Protocols



Internet of Things

Physical Computing

Sensing/Actuating

Messaging

IoT Platforms

Intro

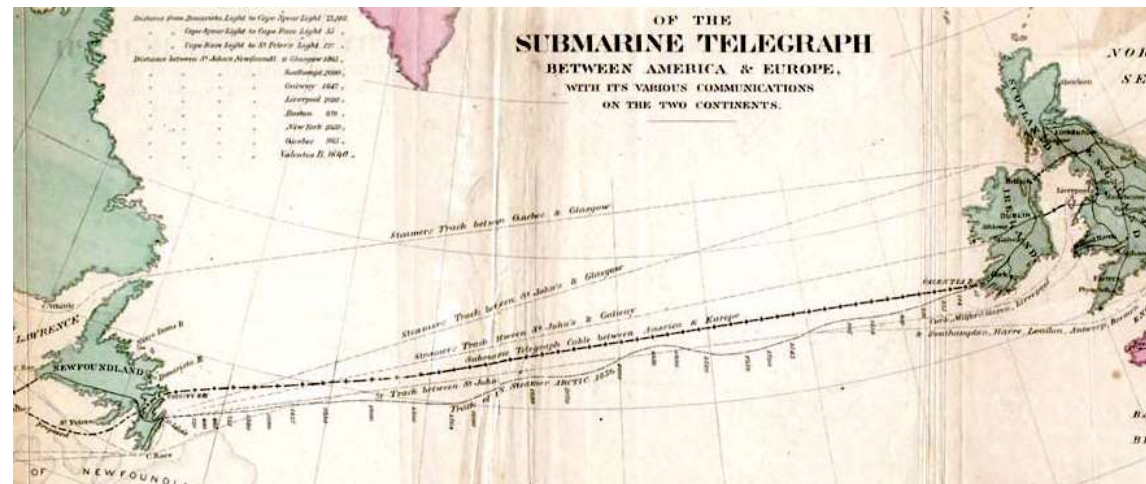
Q1: 1850s - Transatlantic Cable

- The first cable was laid in the 1850s from Valentia Island to Newfoundland
- Messages **encoded** using Morse Code using electrical **signal** through cable.
 - it took 2 minutes to transmit just one character!(Now 30 Tb/s capacity using optical cables(optical signalling))



Encoding is the process of converting data into a specific format or code before transmission.

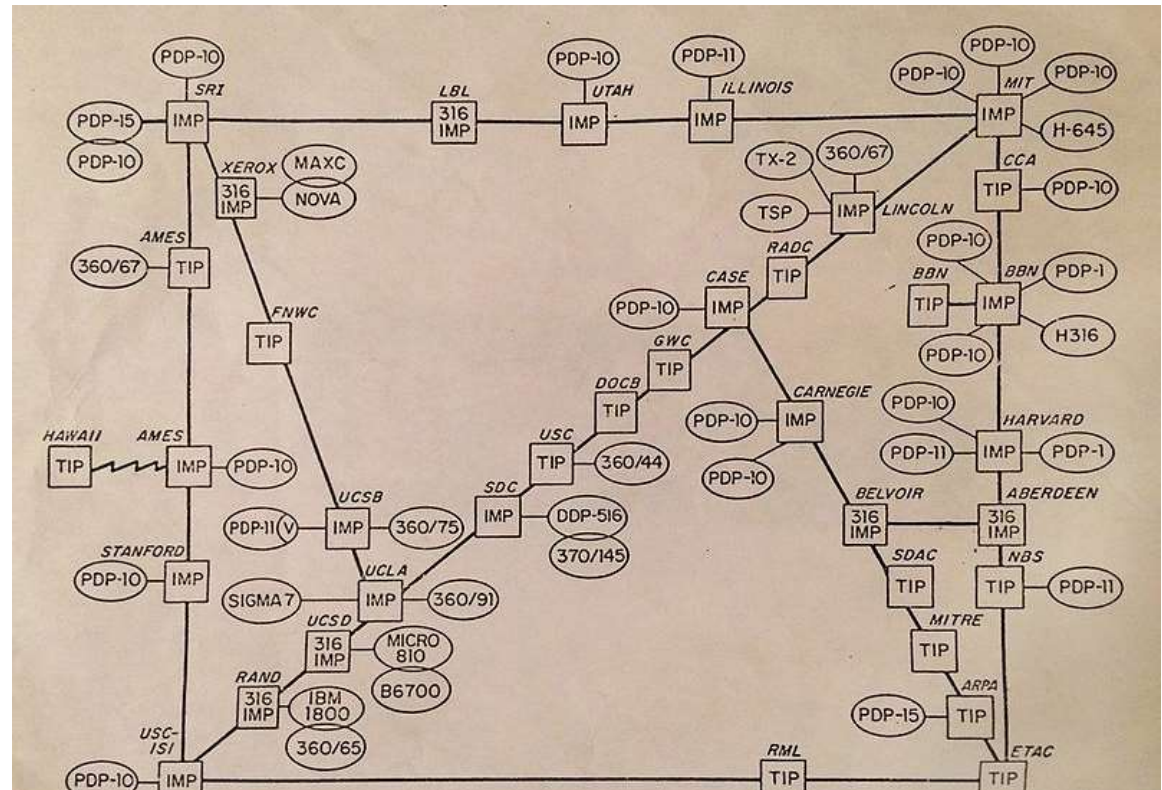
A **Signal** refers to the physical manifestation of data. E.g. electrical voltage, light pulses, or electromagnetic waves



<https://www.telecomtv.com/content/emerging-tech/real-time-trans-atlantic-submarine-cable-capacity-and-wavelength-records-broken-again-40600/>

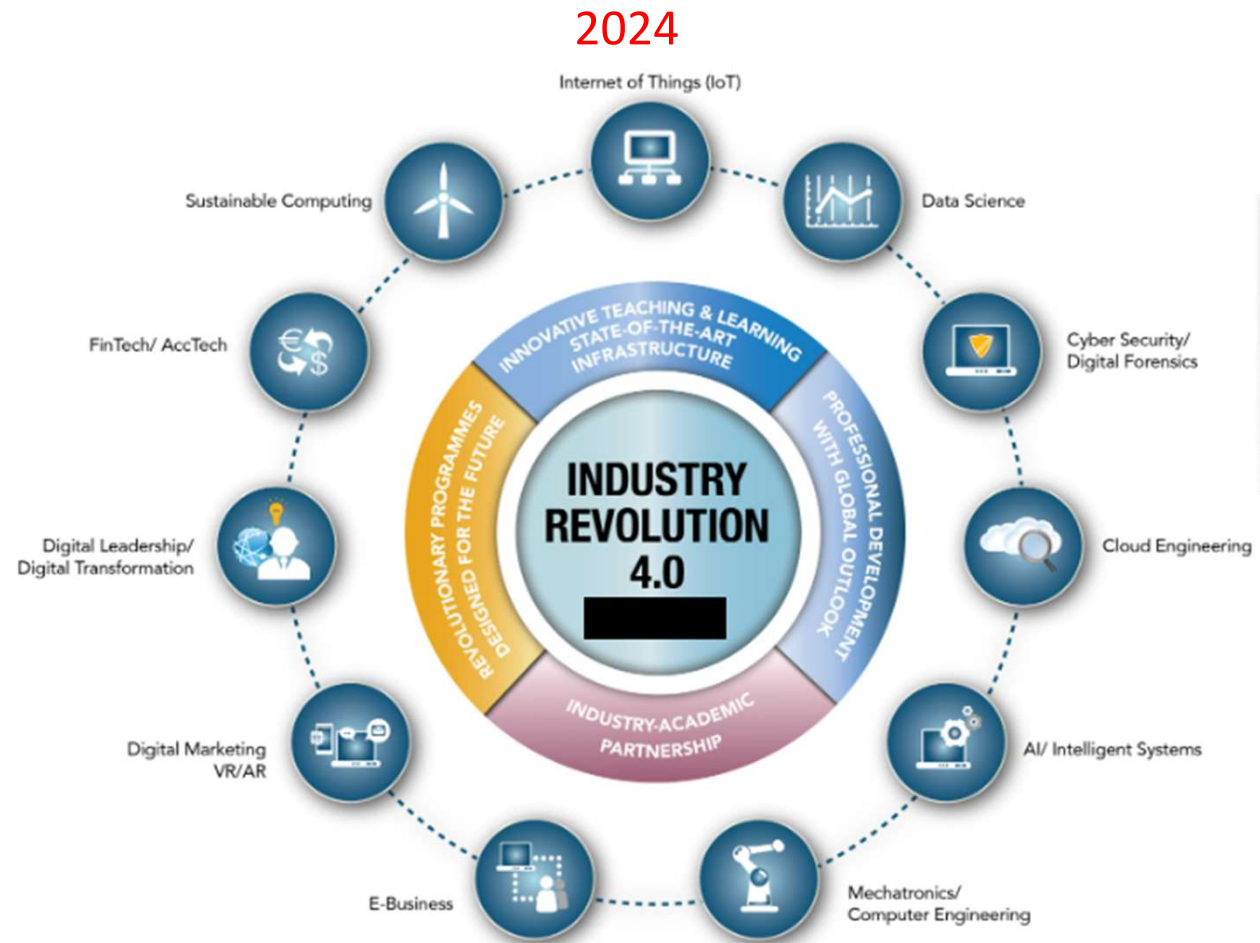
Q1: 1960s - Arpanet

- US government responds to launch of Sputnik by setting up ARPA, the Advanced Research Projects Agency
- **ARPANET** created in 1969 connecting computers in UCLA and Stanford
- In 1973, it was called the "internetwork" or "*internet*" for short
- **The internet is 50 years old**
- **1st Wide Area Packet Switched Network**
 - data is divided into small packets for transmission



Computer Networks

- Devices & Machines are effective/beautiful in isolation...
- Connecting a machine to a network creates amazing possibilities/challenges...
- Key points:
 - Computer networks are no longer only used to connect conventional “computers”
 - Part of everyday life
 - “Wired people should know something about wires”



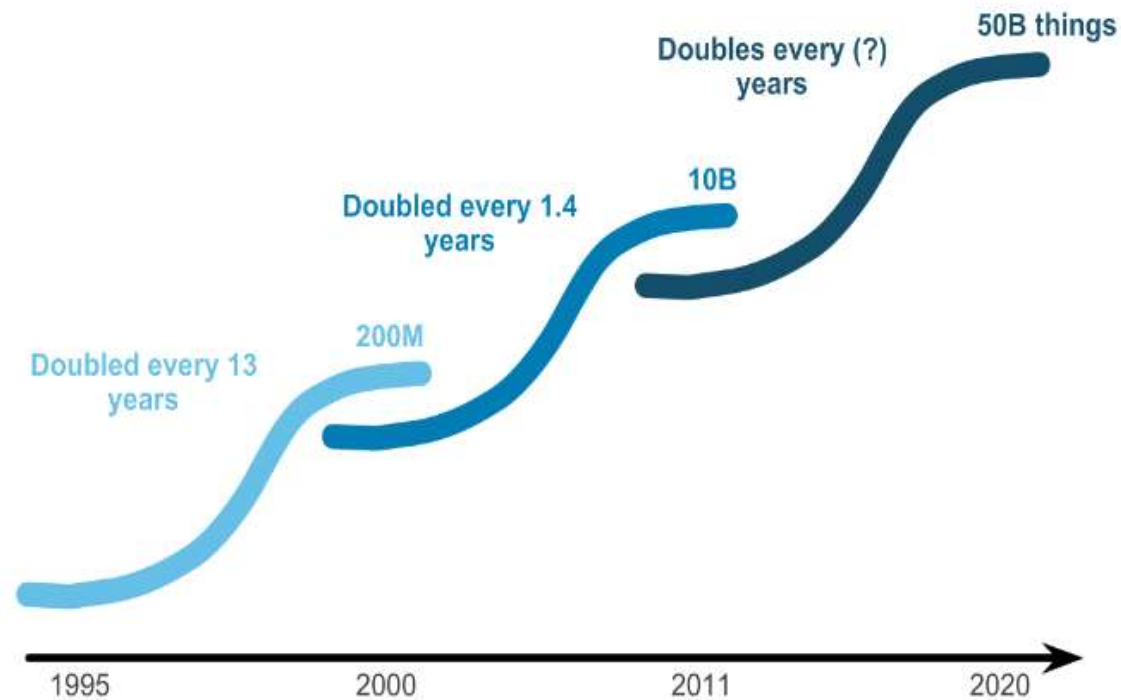
QI: Computer Networks Growth

"Fixed" Computing
(You go to the
device)

Mobility/BYOD
(The device goes
with you)

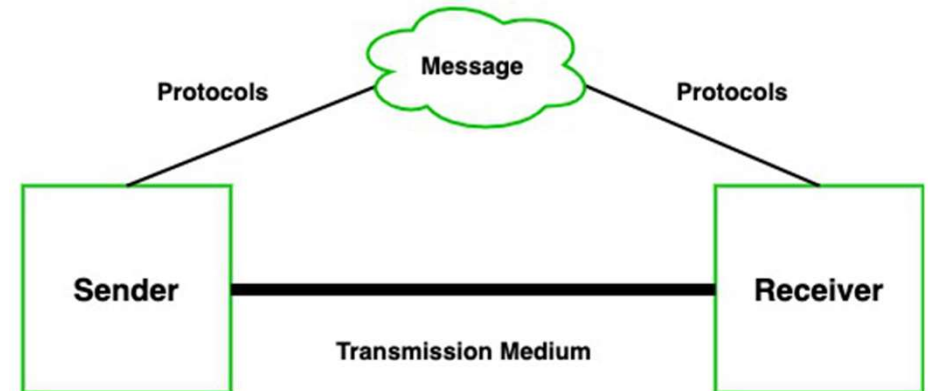
Internet of Things
(Age of Devices)

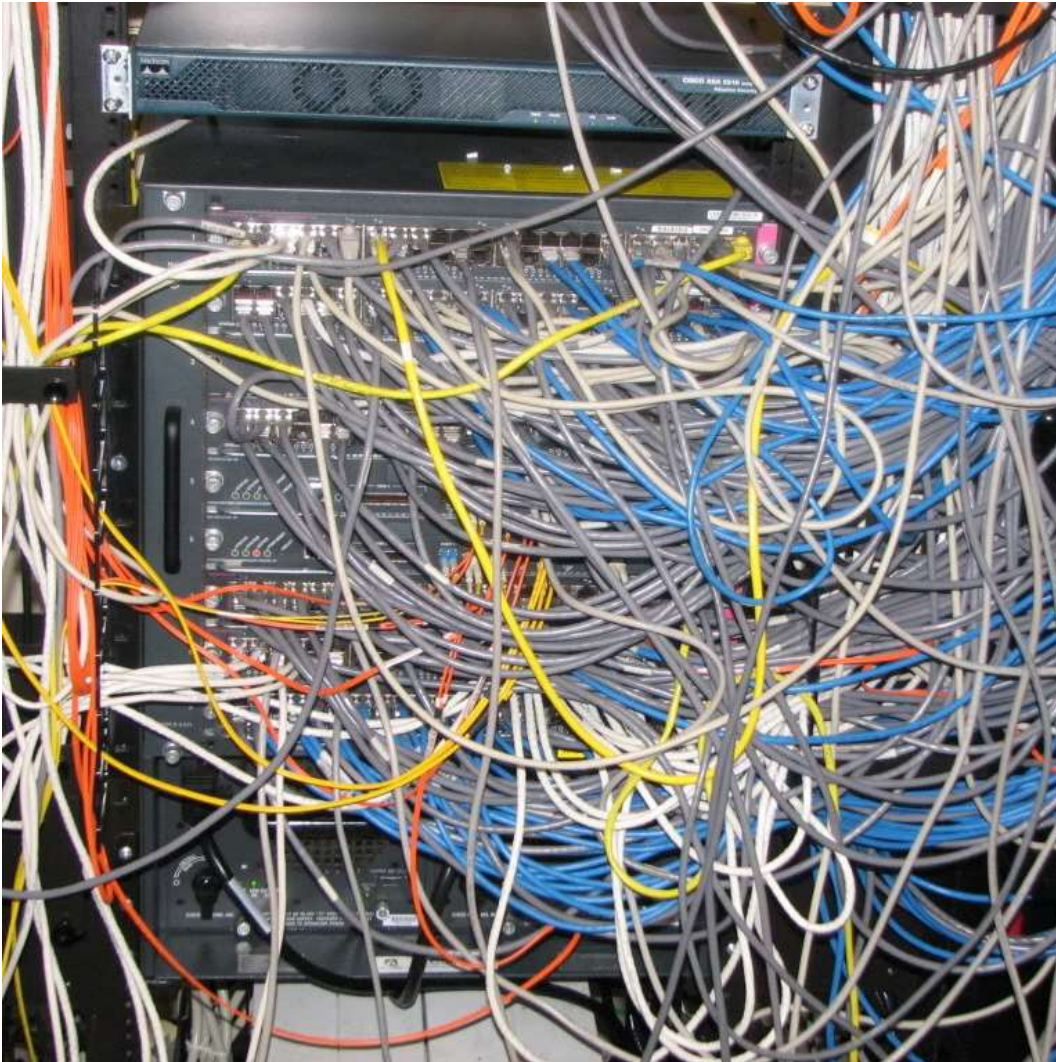
**Internet of
Everything**
(People, Process,
Data, Things)



Networks - Under the bonnet

- Inherently **Physical**:
 - Devices need some form of **physical channel** to communicate
 - Devices need specific hardware to **interface** to that channel (eg. **Antenna**, **adapter**, and associated electronics)

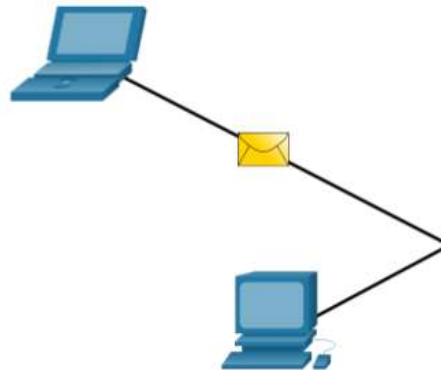
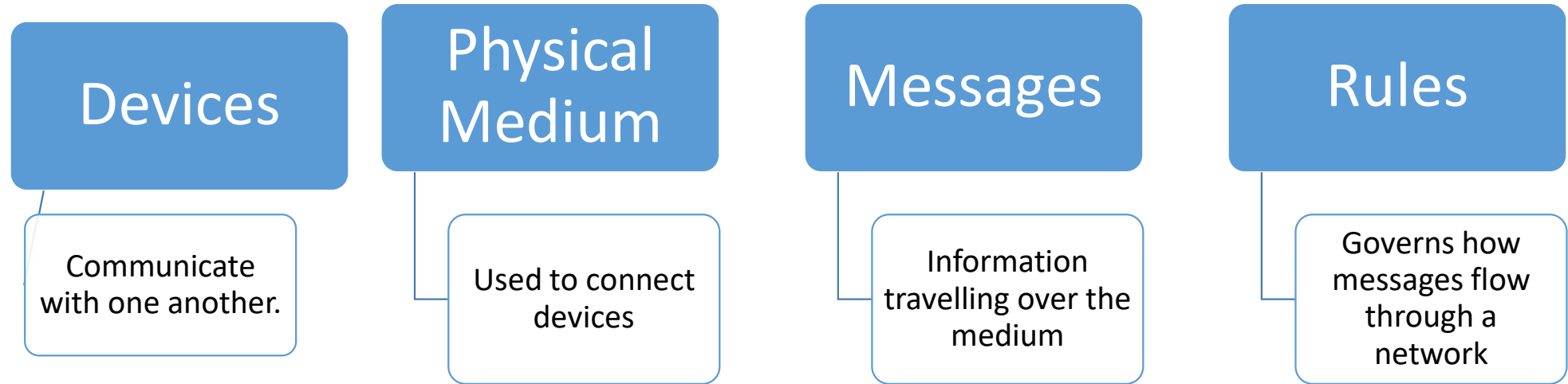




Networks – Under the bonnet

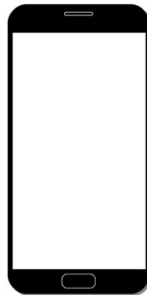
- Once connected, it can get complicated!
- Sophisticated combination of protocols, software, hardware, algorithms, configurations, policies...
- Security, privacy, access, quality of service, wired/wireless...

Components of Computer Networks



Components of Computer Networks - Device Types

- End Devices

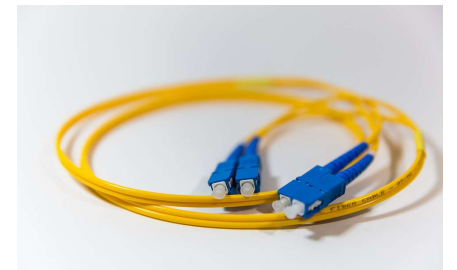
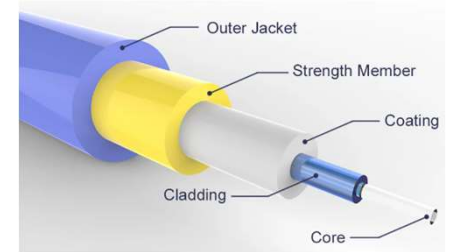
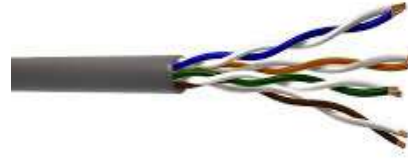


- Infrastructure/intermediary Devices



Components of Computer Networks - Physical Medium

- Wired

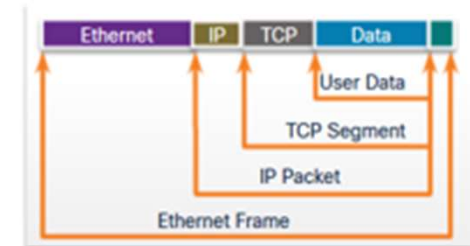
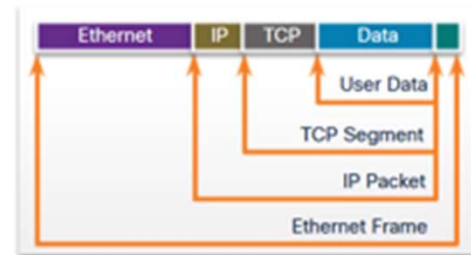
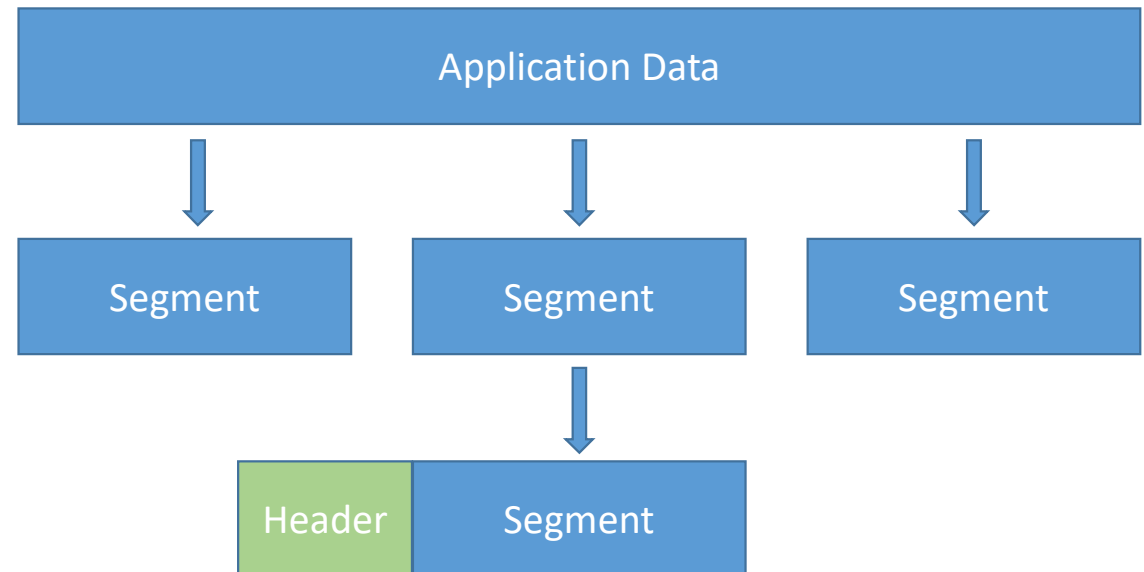


- Wireless



Components of Computer Networks - Messages

- Data is divided into smaller parts during transmission
 - Segmentation
- The benefits of doing so:
 - Many different conversations can be interleaved on the network(multiplexing)
 - Increases reliability of network communications.
 - The separate pieces of each message need not travel the same pathway across the network from source to destination
- Adds complexity however:
 - Addressing, labeling, sending, receiving.
 - Reassembling
- NEED RULES FOR THIS...



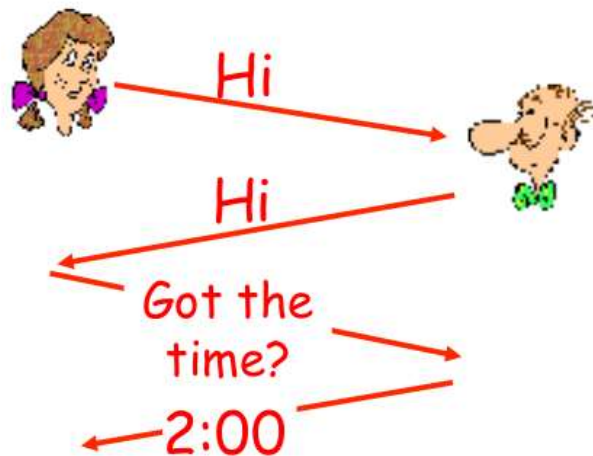
Components of Computer Networks - Rules

- Humans have generally accepted protocols for interaction:
 - Identified sender and receiver
 - Agreed upon method of communicating (face-to-face, telephone...)
 - Common language and grammar
 - Speed and timing of delivery
 - Confirmation or acknowledgement requirements
- All communication activity on the Internet is governed by protocols



Components of Computer Networks - Rules

Human Protocol



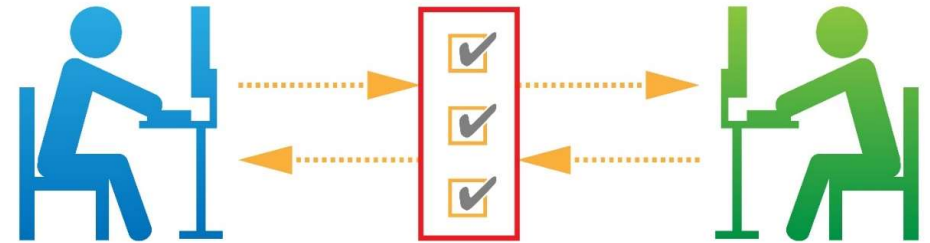
Network Protocol



time
↓

Network Protocols

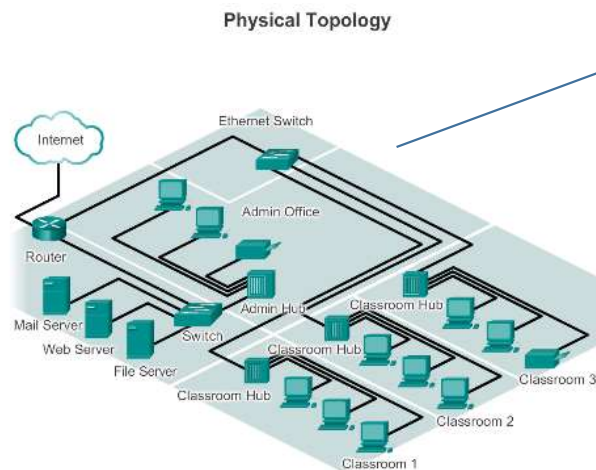
- Machines rather than humans
- All communication activity in Internet governed by protocols
- Devices use agreed-upon protocols to communicate .
- Protocols may have may have one or functions.
- Protocols define
 - Format, order of messages sent and received among network entities
 - Actions taken on msg transmission, receipt



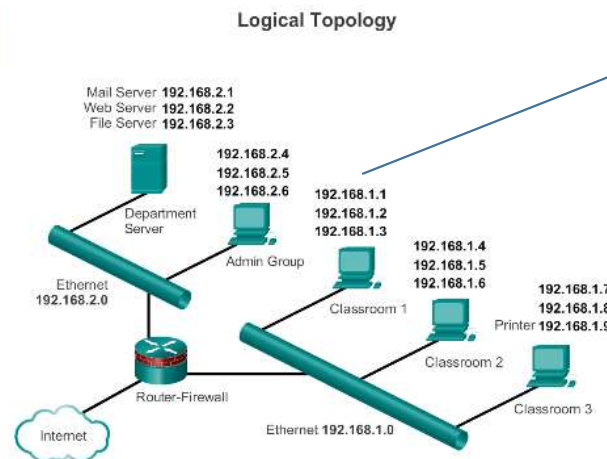
Function	Description
Addressing	Identifies sender and receiver
Reliability	Provides guaranteed delivery
Flow Control	Ensures data flows at an efficient rate
Sequencing	Uniquely labels each transmitted segment of data
Error Detection	Determines if data became corrupted during transmission
Application Interface	Process-to-process communications between network applications

Network Topology and Representations

Network diagrams are called **topology diagrams**. Use accepted symbols to represent devices within the network.

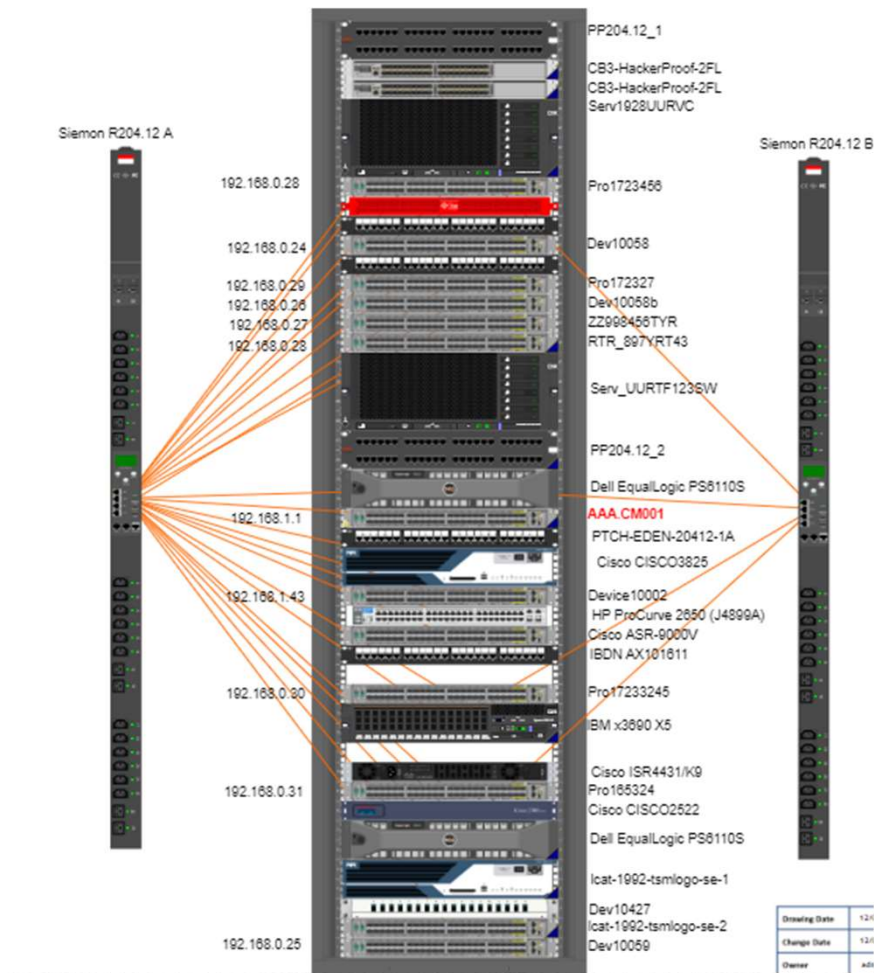


Physical topology diagrams illustrate the physical location of intermediary devices and cable installation.

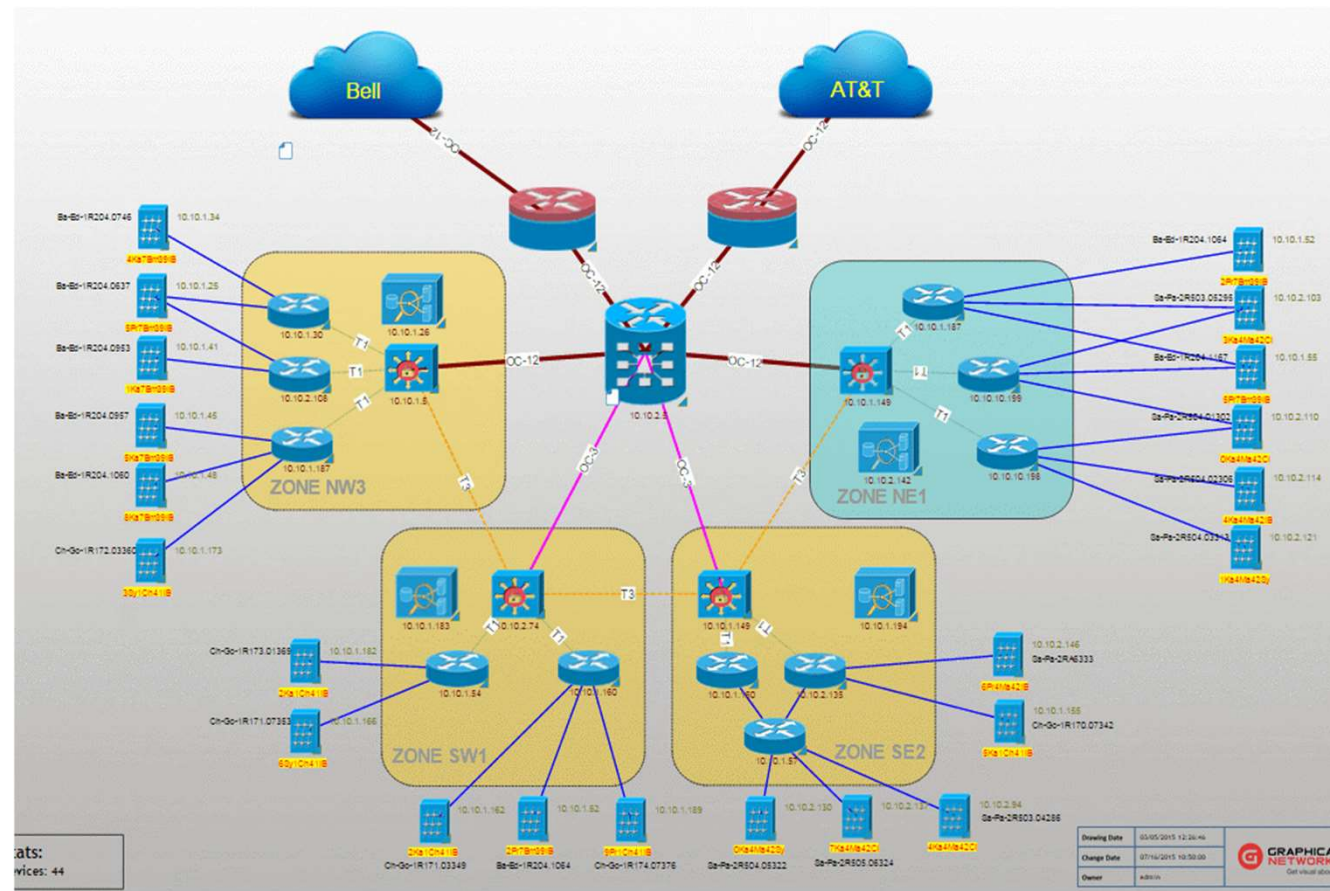


Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.

Network Topologies Examples



Physical topology diagrams



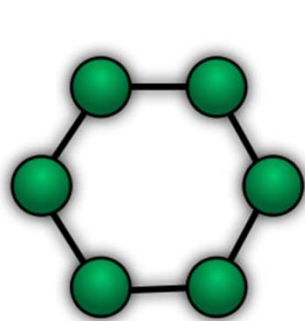
Logical topology diagrams

Network Topology

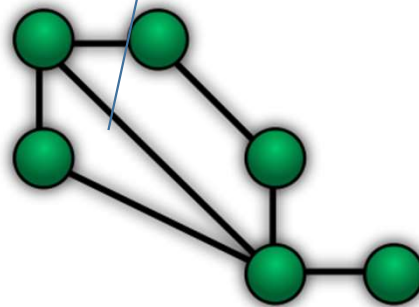
What connection to use?
Unorganised....

Centre is single point of failure

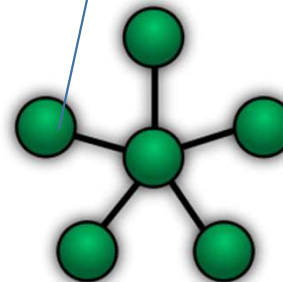
Will not scale well



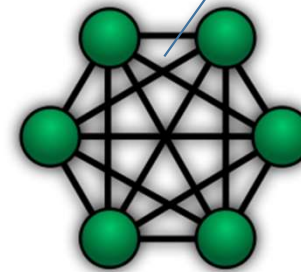
Ring



Mesh



Star

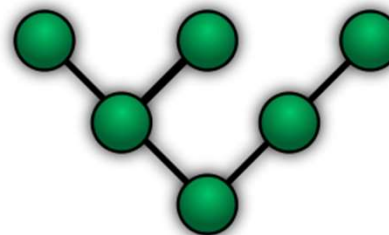


Fully Connected



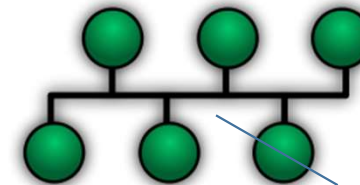
Line

What if one device goes down???



Tree

source: wikipedia



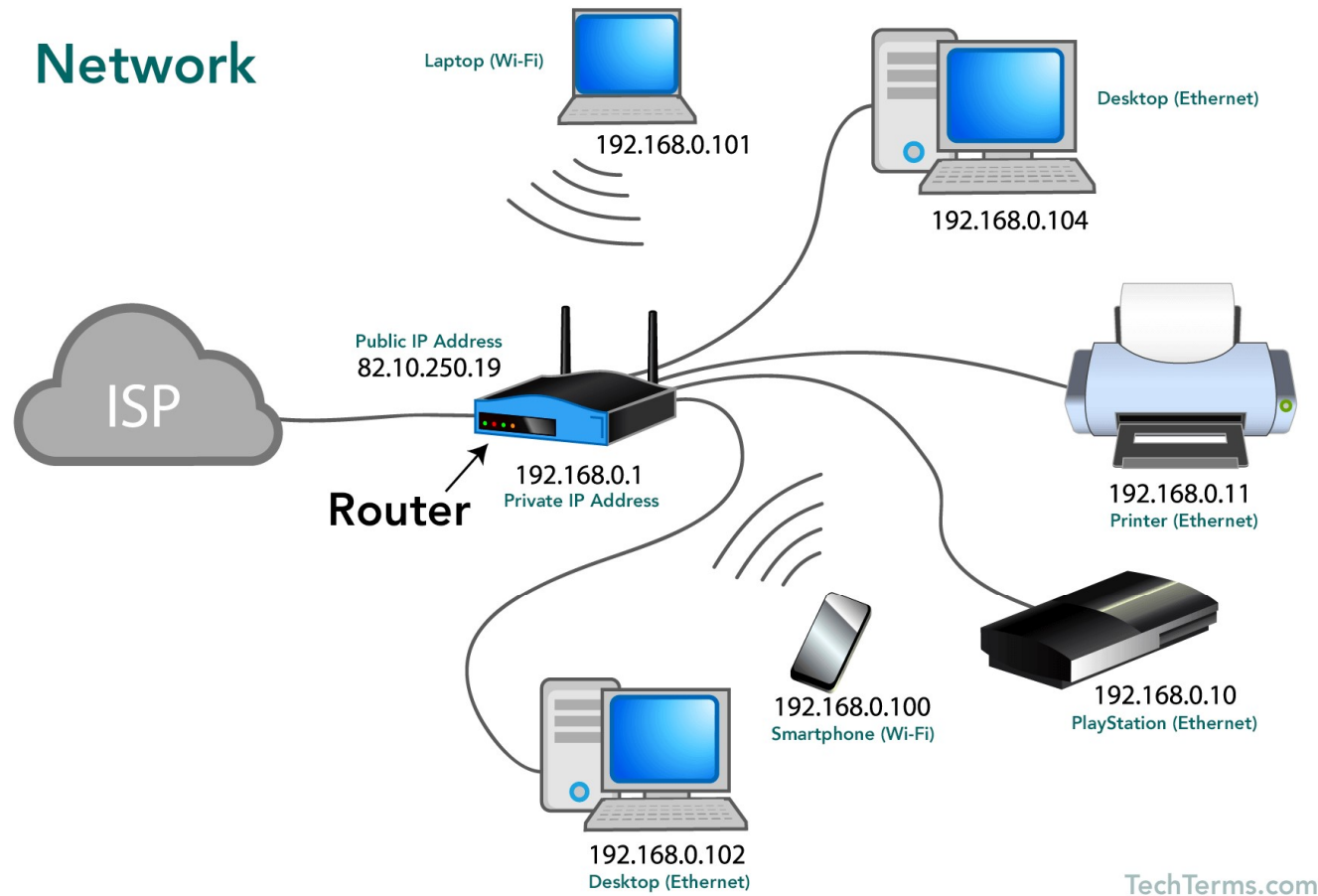
Bus

2 devices communicate simultaneously??
?

Network Types

- Small Home Network
 - A few devices connected to each other and the Internet
- Small Office/Home Office
 - Connects to corporate network
 - More complex intermediary devices/design(VPN/Firewalls/VLAN)
- Medium to Large Network
 - Many locations/networks. >100 devices
- Worldwide Networks
 - Millions of connected devices via interconnected networks across the globe.
 - E.g. the Internet

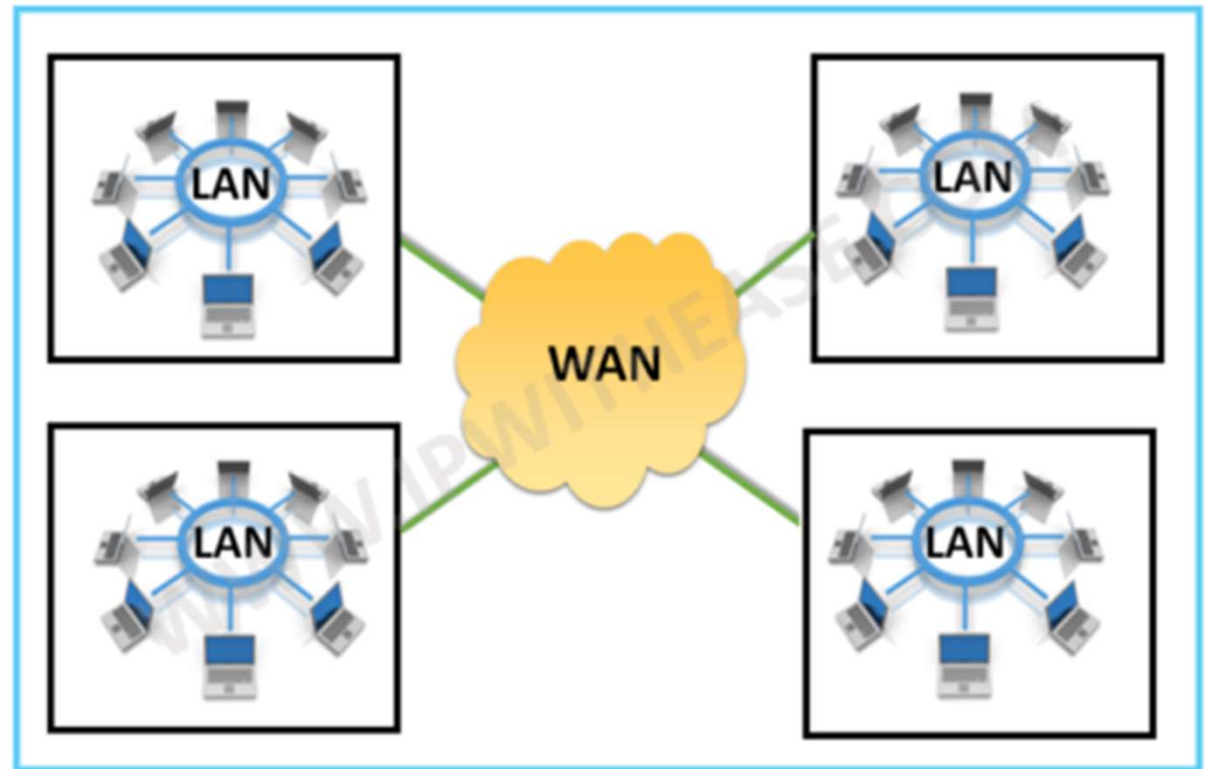
Network Types - Typical Home Network



Source: <https://techterms.com/definition/network>

Network Types -

- LAN(local area network)
 - interconnects computers within a limited area, such as a home, office building, school, or campus.
 - Administered by single person or organisation
- WAN(wide area network)
 - covers a broad area.
 - Interconnects LANs
- Internet
 - Global collection of interconnected LANS and WANS
 - No ownership. Groups formed to maintain structure:
 - IETF
 - ICANN

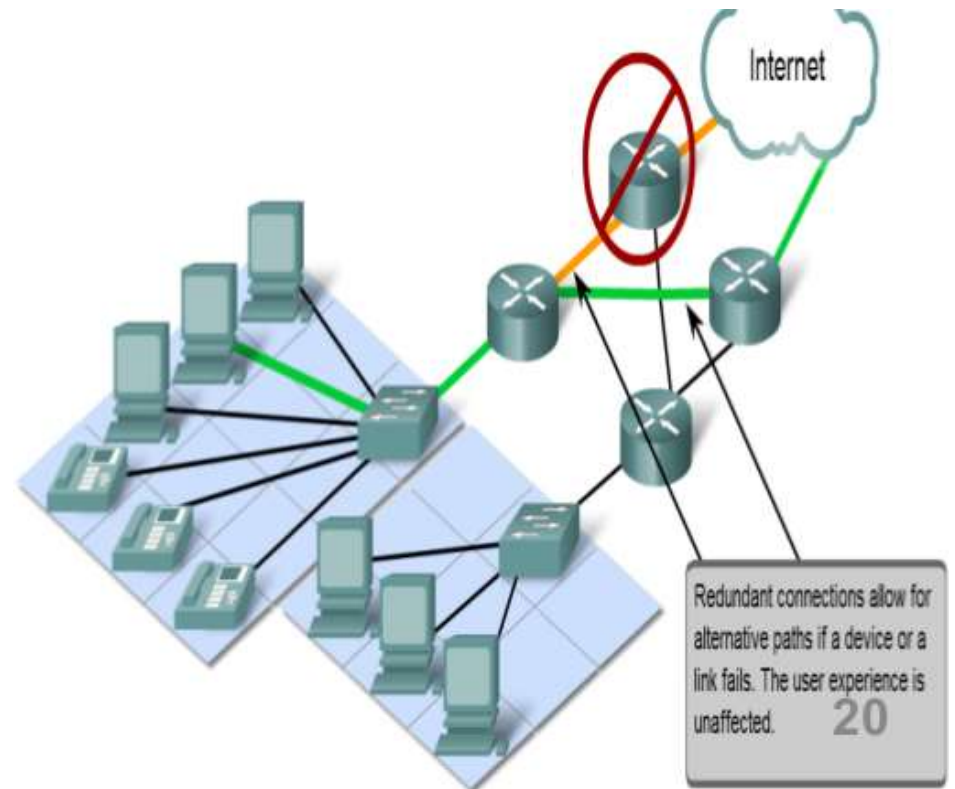


Some Networking Vocabulary

- **Network Interface/Adapter:** any kind of software interface to networking hardware. (e.g. wifi interface and wired interface). A network interface may be associated with a physical device, or it may be a representation of a virtual interface. (e.g. interfaces on your virtual machines)
- **LAN:** Local Area Network refers to a network or a portion of a network that is not publicly accessible to the greater internet.
- **WAN:** Wide Area Network. Much Larger and extensive than a LAN. Often used to refer to the Internet, as a whole.
- **Node:** General term usually for a device on a network. Every node has a unique network address.
- **Media Access Control(MAC) address:** Used to distinguish specific devices. A unique address that each device is assigned during manufacturing. Used to differentiate it from every other device on the internet. Typically, each network interface has a MAC address.
- **IP:** protocols that allow the internet to work. IP addresses are unique on each network and they allow machines to address each other across networks.

Reliable Networks - Network Characteristics

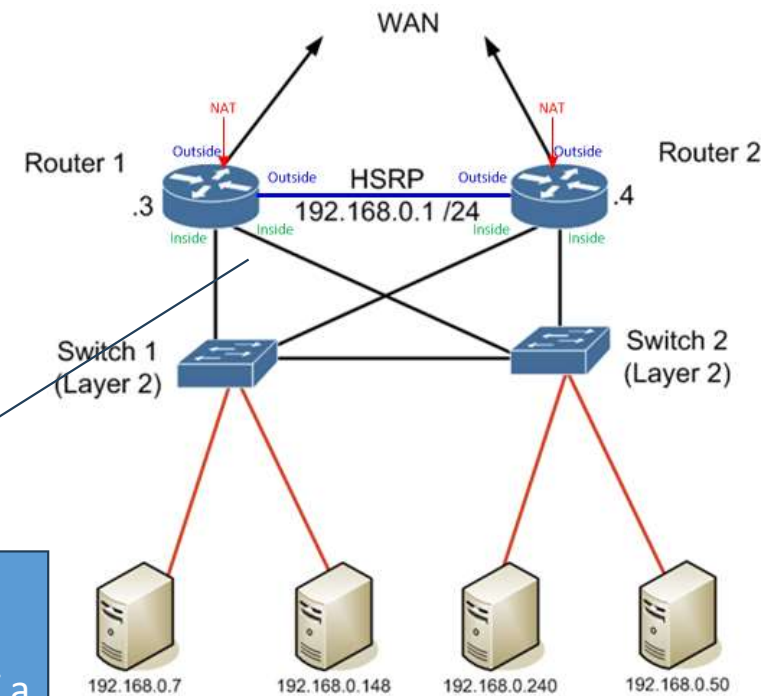
- Network architecture refers to:
 - Design of a Network
 - network's physical components and its organisation
 - the technologies that support the infrastructure
- 4 general characteristics to meet user expectations
 - Fault tolerance
 - Scalability
 - Quality of service (QoS)
 - Security



Reliable Networks – Fault Tolerance

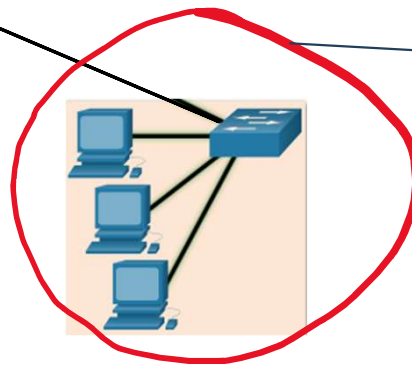
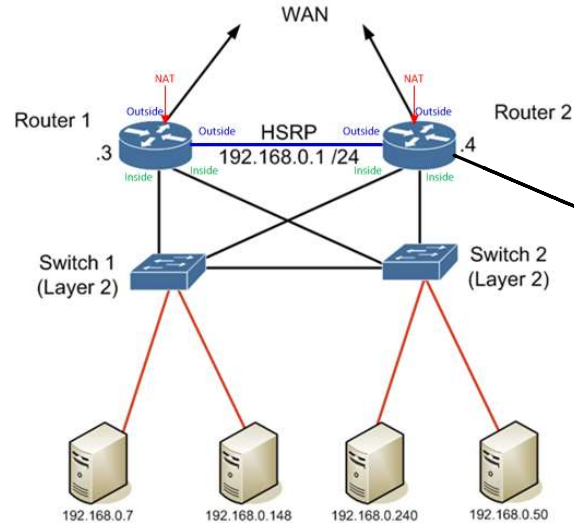
- Refers to limiting impact failures
- Implemented using **Packet Switched Networks** and **Multiple Paths**
 - Packet switching splits transmitted data into packets that are routed individually over a network.
 - Packets can take different routes from source to destination
- Not possible with **Circuit Switched Networks** (eg old telecoms/phone networks)

Redundant connections provide alternative paths if a device or link fails



Reliable Networks - Scalability

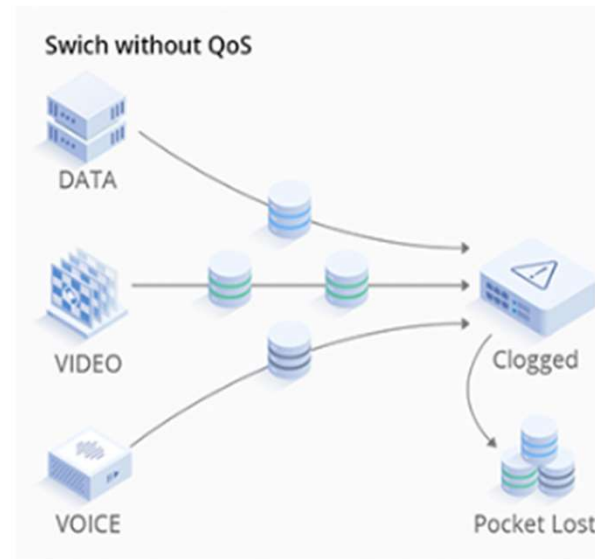
- Networks should be able to scale(grow) easily
- Performance should not be affected for existing users
- Standards/protocols help to achieve this.



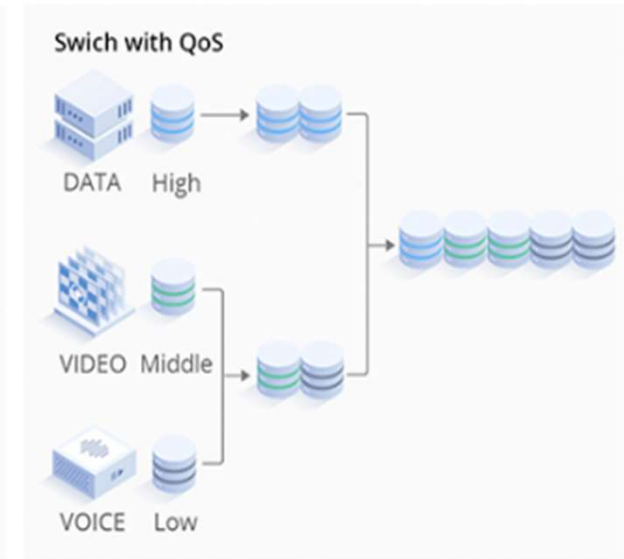
Adding new users/networks
without degrading
performance

Reliable Networks - QoS

- Ever watched laggy/freezing video
 - Caused by higher demand for Bandwidth available
- Certain traffic types require higher performance/bandwidth
- Quality of Service used to do this
 - QoS policy enables routers to manage flow of media traffic(voice/video)
- Example: Routers prioritise voice/video traffic over web page traffic



Uncontrolled traffic causes network congestion and packet loss



Classify and prioritize traffic for optimum efficiency

Reliable Networks - Network Security



- Network infrastructure security
 - Preventing unauthorized access to the physical devices
- Information Security
 - Security of data transmitted on network
- Confidentiality – only intended recipients can read the data
- Integrity – assurance that the data has not be altered with during transmission
- Availability – assurance of timely and reliable access to data for authorised users



Key Points so far

- Networks are everywhere
- 4 Components of every network
 - Devices
 - Medium
 - Protocols (Rules)
 - Messages (Data)
- Networks are connecting everything (not just for PCs/Laptops)
- Networks have a Topology
- Some key characteristics of a network

Internet of Things

- Internet → The worldwide network of interconnected computer networks, based on a standard communication protocol (TCP/IP).
- Thing → An object not precisely identifiable.
- Internet of Things (IoT) → A worldwide network of interconnected objects uniquely addressable, based on standard communication protocol.



What's the difference...

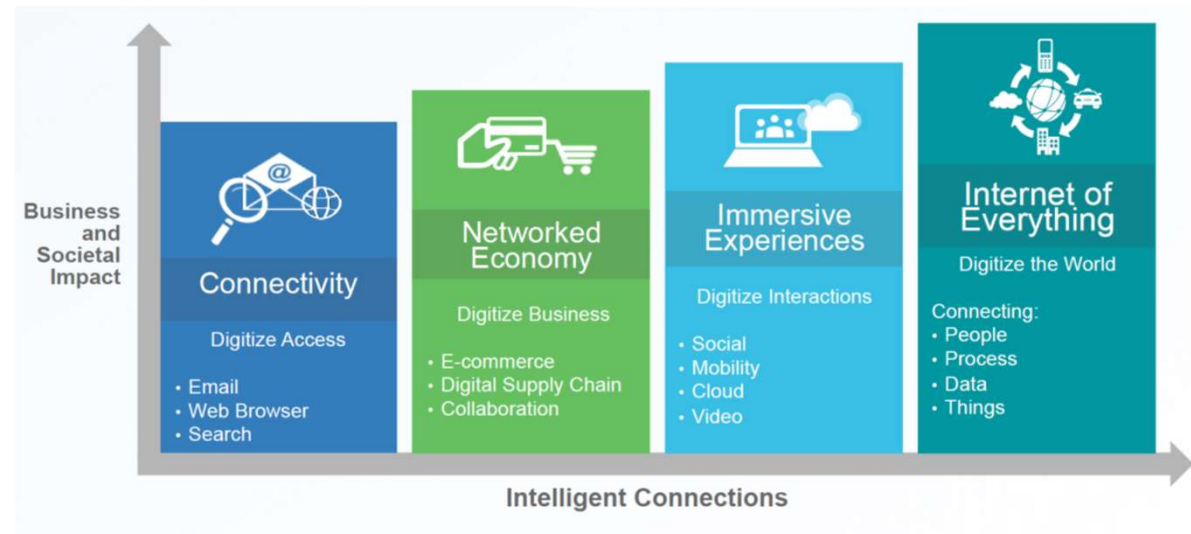
- Extending the current Internet and providing connection, communication, and inter-networking between devices and physical objects, or "Things," (even biological things!)



<https://www.capitatraininginterpreting.com/the-internet-of-things/>

IoT Evolution

- Started with **connectivity** among people for sharing information.
- Led to a “flat-world” where **everyone** across the world is connected.
- Advancement in **cloud computing** and immersive experience led towards universal accessibility of data.
- Combination of immersive experiences, connectivity and advancement in electronics further leading to **Internet of Everything (IoE)**

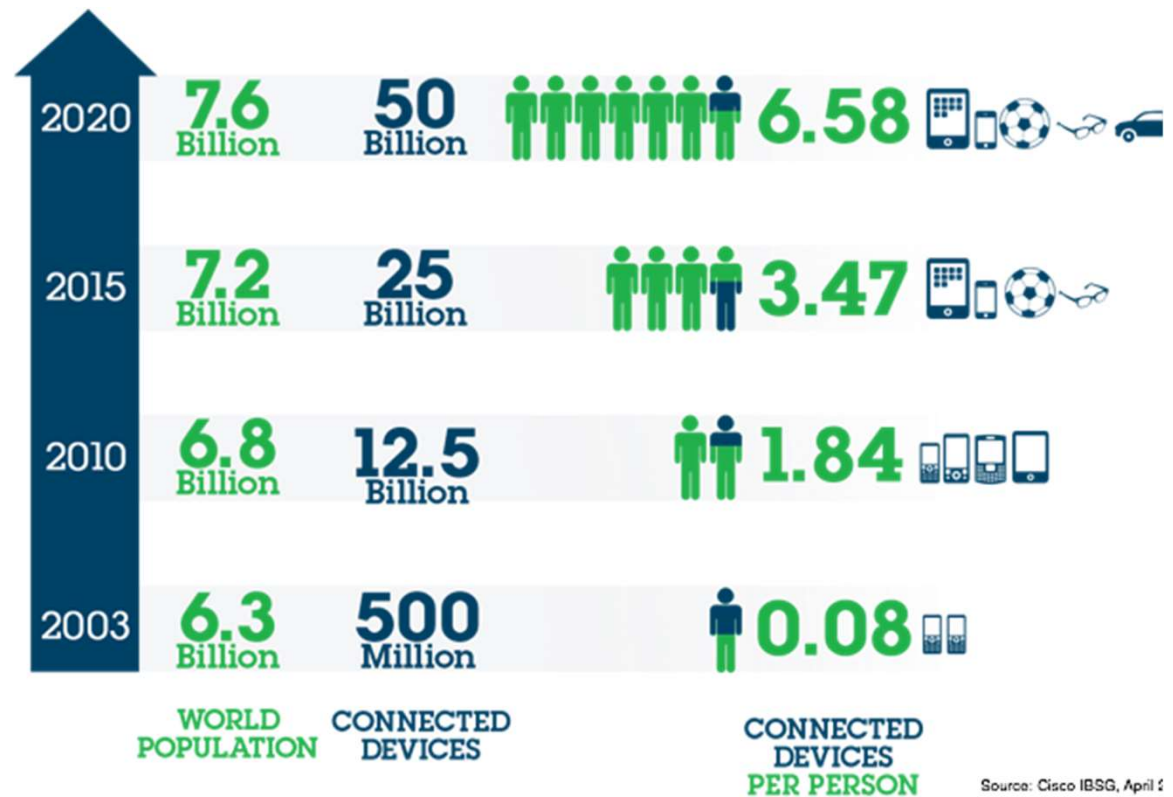


Source: Cisco

From an estimated 8.4 billion devices in 2017, the number of connected IoT devices is projected to reach over 25 billion by 2030

IoT Potential

- Conservatively → 20 billion newly connected devices will be deployed.
- First public website went live at CERN in 1990.
 - It took 15 years to reach 1 billion people on earth over the internet.
 - IoT is looking to add 6 billion connected devices per year.
- Economic impact
 - New revenue streams
 - Reducing costs
 - Reducing time to market
 - Improving supply chain
 - Reducing production loss
 - Increasing productivity



Networking and IoT

- The Roles of Networks
 - Managing connected devices (discovery, join, leave, etc).
 - Relaying data packets from the source to the destination node in the network.
- IoT is a distributed system. All nodes need to perform networking related tasks.
- Main concerns as before: Reliability, Performance, Security, QOS, Scalability



Learning about Networks/IoT

- Can't create complex networks at home however you can simulate "virtual networks"
 - Virtualbox hypervisor
 - Packet Tracer
- Can use programmable, multi-channel, prototyping device to investigate different mediums, protocols, IoT etc.
 - Raspberry Pi has bluetooth, Wifi, Ethernet, SPI, I2C...

